DOMAIN-SPECIFIC-WORK-OF-DATA-ANALYST-SCIENTIST

(Insurance)

01. Overview — What is this domain?

Insurance is fundamentally a business of **pricing and pooling risk**. Companies collect premiums from many policyholders to pay out claims for the few who suffer a covered loss, and the entire business model depends on **accurately estimating the probability and cost of future events**. This domain includes:

- **Life Insurance** – mortality risk, term/whole life products, annuities
- **Health Insurance** – medical claims, hospitalization, wellness programs
- **General/Non-Life Insurance** – motor, home, travel, marine, fire insurance
- **Property & Casualty (P&C)** – covers damage to property and liability claims
- **Reinsurance** – insurance for insurance companies, spreading catastrophic risk
- **InsurTech** – digital-first, usage-based, and on-demand insurance models

Why data analytics/science matters here

1. **Pricing is fundamentally statistical** — actuaries and data scientists estimate risk using historical loss data, demographics, and behavior.
2. **Claims are the biggest cost driver** — fraud detection and claims efficiency directly affect profitability (the "combined ratio").
3. **Underwriting decisions must scale** — millions of policies need consistent, data-driven risk assessment.
4. **Customer retention is data-driven** — churn prediction and personalized pricing keep policyholders from lapsing to competitors.
5. **Heavy regulation** — solvency, reserving, and fair pricing practices are all governed by regulators.

Industry context & key regulatory bodies

Region	Regulator(s)
India	IRDAI (Insurance Regulatory and Development Authority of India)
USA	State Insurance Departments, NAIC (National Association of Insurance Commissioners)
UK/EU	PRA, FCA, EIOPA
Global	IAIS (International Association of Insurance Supervisors)

Key frameworks:

- **Solvency II (EU)** – capital adequacy requirements for insurers
- **IFRS 17** – new global accounting standard for insurance contracts
- **IRDAI Guidelines (India)** – product approval, claims settlement ratio disclosures
- **Actuarial Standards of Practice** – governs reserving and pricing methodology

Typical business functions a Data Analyst/Scientist supports

- Underwriting & Risk Pricing
- Claims Processing & Fraud Detection
- Customer Retention & Cross-sell/Up-sell
- Actuarial Reserving & Reporting
- Reinsurance Optimization
- Distribution/Agent Channel Analytics

Why this domain is attractive for Data Analysts/Scientists

- Rich historical data on risk events spanning decades
- Strong overlap with actuarial science — a well-established, respected quantitative discipline
- High-impact fraud detection work with direct P&L impact
- Career path: Analyst → Senior Analyst/Actuarial Analyst → Data Scientist → Chief Actuary/Chief Risk Officer

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02. Role of a Data Analyst/Scientist — Day-to-Day & Full Project Lifecycle

A. Day-to-Day Responsibilities (Snapshot)

- Extracting and validating policy/claims data from core insurance systems
- Building/maintaining pricing models and rating algorithms
- Monitoring claims fraud detection dashboards
- Running loss ratio and combined ratio reports by product/segment
- Supporting actuarial reserving calculations with data pulls and validation
- Building churn/retention models for renewal season
- Presenting risk segment findings to underwriting and product teams

B. End-to-End Project Lifecycle (Example: Motor Insurance Claims Fraud Detection)

Step 1: Business Problem Understanding

- Meet with the Claims Head/Fraud Investigation Unit to understand the pain point.
    - Example: *"Fraudulent claims are estimated at 8% of total motor claims payout, but our investigators can only manually review 15% of claims."*
- Translate into an analytical framing:
    - Business ask → *Reduce fraud leakage without slowing down genuine claims*
    - Analytical framing → *Build a fraud scoring model to prioritize claims for investigation at First Notice of Loss (FNOL)*
- Define success metrics: fraud detection rate, investigation cost savings, false positive rate (to avoid harassing genuine customers)

Step 2: Data Collection & Understanding

- Identify sources: policy database, claims history, FNOL call transcripts/notes, garage/repair estimates, past investigation outcomes (confirmed fraud vs genuine)

- Understand claim lifecycle stages (FNOL → investigation → settlement → closure)
- Define target variable clearly: Fraud = 1 if claim was confirmed fraudulent by the Special Investigation Unit (SIU); else Genuine = 0
- Check historical labeling quality — investigated cases are a biased sample (only suspicious claims get investigated), which needs to be accounted for in modeling

Step 3: Data Extraction

- Pull policy attributes (vehicle type, policy tenure, previous claims, no-claim bonus history)
- Pull claim attributes (claim amount, time since policy inception, day/time of incident, location, garage used)
- Join with external data where available (vehicle registration databases, past claim history across insurers via industry fraud bureaus)

Step 4: Data Cleaning & Preprocessing

- Handle missing values in claim narrative fields, inconsistent garage/repair codes
- Standardize location data (pin codes, city names) for geographic risk analysis
- Handle severe class imbalance (confirmed fraud is often <5% of all claims)
- Deduplicate claims linked to the same incident (common in multi-party accidents)

Step 5: Exploratory Data Analysis (EDA)

- Analyze fraud rate by claim type, time since policy start (claims filed suspiciously soon after purchase are higher risk), claim amount distribution
- Identify red-flag patterns: claims filed just before policy expiry, unusually high repair estimates, multiple past claims by same policyholder
- Geographic hotspot analysis — fraud rings often cluster in specific regions/garage networks

Step 6: Feature Engineering

- Create derived features:
    - **Days since policy inception at time of claim**
    - **Number of previous claims by policyholder**

    - **Claim amount vs vehicle's insured value ratio**
    - **Garage risk score** (based on historical fraud rate associated with that garage)
    - **Network features** – shared phone numbers/addresses/garages across multiple claims (graph-based features)
- Encode categorical variables (vehicle type, claim type, region)

Step 7: Model Building

- Split data into train/test, being careful about the biased-sample issue (consider techniques like semi-supervised learning if only investigated claims are labeled)
- Common algorithms:
    - **Logistic Regression** – interpretable baseline, often required for regulatory/audit purposes
    - **Random Forest / XGBoost / LightGBM** – for higher predictive accuracy
    - **Anomaly Detection (Isolation Forest, Autoencoders)** – to catch novel fraud patterns not well-represented in historical labeled data
    - **Graph-based Models** – to detect organized fraud rings via shared entities across claims
- Combine model score with business rules (e.g., automatic flag if claim amount exceeds a threshold within X days of policy start)

Step 8: Model Validation

- Evaluate using Precision/Recall (Recall is critical — missing fraud is costly, but so are false positives that annoy genuine customers)
- Use **Precision-Recall AUC** rather than plain ROC-AUC, since fraud is a rare-event problem
- Validate on a holdout time period (claims from a later quarter) to simulate real-world deployment
- Review model output with SIU investigators for face validity — do the flagged claims make business sense?

Step 9: Model Governance & Documentation

- Document model logic, feature list, and limitations for actuarial/compliance review
- Ensure model doesn't introduce discriminatory bias against any protected demographic (fair treatment of customers is a regulatory expectation)
- Get sign-off from Risk/Compliance before deployment

Step 10: Deployment

- Integrate fraud score into the claims processing system at FNOL stage
- Set up a tiered investigation workflow: high-score claims → mandatory SIU review; medium-score → sampling-based review; low-score → fast-tracked settlement
- Run in shadow mode initially (model scores generated but not yet driving decisions) to validate against investigator judgment

Step 11: Monitoring & Maintenance

- Track fraud detection rate, false positive rate, and investigation cost savings monthly
- Monitor for fraud pattern drift (fraudsters adapt over time, so models need periodic retraining)
- Refresh model with newly confirmed fraud/genuine outcomes on a regular cycle (e.g., every 6 months)

Step 12: Reporting & Stakeholder Communication

- Present impact metrics (fraud leakage reduction, cost savings, ROI of the fraud detection program) to senior management
- Provide feedback loops to underwriting teams on newly discovered risk patterns (e.g., certain garages or regions needing tighter scrutiny)

C. This Lifecycle Applies (with Variations) to Other Insurance Use Cases

- **Pricing/Underwriting Models** – similar steps but target variable is claim frequency/severity rather than fraud, output feeds into premium calculation
- **Churn/Retention Models** – target variable is policy lapse/non-renewal instead of fraud
- **Reserving Models (Actuarial)** – focuses on estimating future claim liabilities rather than classification, heavier use of actuarial triangulation methods

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03. Key Metrics & KPIs

Underwriting & Pricing Metrics

Metric	Meaning
Loss Ratio	Claims paid ÷ Premiums earned — core profitability indicator
Combined Ratio	(Losses + Expenses) ÷ Premiums earned — below 100% means underwriting profit
Claim Frequency	Number of claims per policy/exposure unit
Claim Severity	Average cost per claim
Loss Cost	Frequency × Severity — expected loss per unit of exposure
Premium Growth Rate	Year-over-year change in written premiums

Claims Metrics

Metric	Meaning
Claims Settlement Ratio	% of claims settled by the insurer (key trust/regulatory metric in India)
Average Claim Turnaround Time (TAT)	Time from claim intimation to settlement
Fraud Detection Rate	% of actual fraudulent claims correctly identified
Straight-Through Processing (STP) Rate	% of claims processed automatically without manual intervention

Customer Metrics

Metric	Meaning
Persistency Ratio	% of policyholders renewing their policy (critical in life insurance)
Policy Lapse Rate	% of policies that lapse due to non-payment/non-renewal
Cross-sell Ratio	Average number of policies held per customer
Customer Lifetime Value (CLV)	Total expected profit from a customer relationship

Actuarial & Solvency Metrics

Metric	Meaning
Solvency Ratio	Available capital relative to required capital — regulatory health measure
IBNR (Incurred But Not Reported) Reserves	Estimated liability for claims that have occurred but not yet been reported
Reserve Adequacy	Whether reserves are sufficient to cover future claim payouts
Expense Ratio	Operating expenses as a % of premiums

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04. Common Datasets & Data Sources

Internal Data Sources

Source System	**Typical Data**
Policy Administration System (PAS)	Policy details, premium amounts, coverage terms, renewal history
Claims Management System	Claim details, incident description, settlement amount, status
Underwriting System	Risk assessment inputs, medical/vehicle inspection reports
CRM System	Customer interactions, complaints, service requests
Agent/Broker Management System	Distribution channel performance, commission data

External Data Sources

Source	**Typical Data**
Industry Fraud Bureaus (e.g., IIB in India)	Cross-insurer claim history to detect multi-insurer fraud
Vehicle Registration Databases	Ownership history, accident records (motor insurance)
Medical Records/Hospital Networks	Claim verification data (health insurance)
Credit Bureau Data	Used in some underwriting models as a proxy risk indicator
Weather/Catastrophe Data	Used for property insurance risk modeling (floods, earthquakes)
Telematics Data	Usage-based insurance — driving behavior from vehicle sensors/apps

Typical Fields/Attributes

- **Policy Data**: sum insured, premium, policy term, product type, rider/add-ons
- **Claims Data**: claim date, claim type, claim amount, cause of loss, settlement date
- **Customer Data**: age, occupation, medical history (health/life), vehicle details (motor)
- **Actuarial Data**: mortality/morbidity tables, loss development triangles

Data Characteristics Specific to Insurance

- **Long-tail claims** — some claims (especially liability/health) can take years to fully develop and settle
- **Highly sensitive data** — medical and financial information requires strict privacy handling
- **Rare-event modeling** — catastrophic losses and fraud are inherently imbalanced classification problems
- **Actuarial data structures** — loss triangles and cohort-based data are unique to this domain

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05. Tools & Technologies

Programming Languages

- **Python** – Pandas, Scikit-learn, XGBoost for modeling and analysis
- **R** – widely used by actuarial teams for statistical/GLM modeling
- **SQL** – for querying policy/claims databases
- **SAS** – still common in large, traditional insurers for actuarial and risk modeling

Actuarial & Pricing Software

- **GLM-based pricing tools** (Emblem, Radar by Willis Towers Watson) – industry-standard for building rating algorithms
- **Prophet (Actuarial software, not Facebook Prophet)** – used for life insurance actuarial modeling
- **ResQ, Arius** – claims reserving software

Visualization & BI Tools

- **Power BI, Tableau** – for claims/underwriting dashboards
- **Excel** – heavily used for actuarial calculations and ad-hoc reporting

Fraud Detection & Advanced Analytics

- **SAS Fraud Management** – enterprise fraud detection platform
- **Graph databases (Neo4j)** – for fraud ring/network detection
- **Python ML libraries** – Scikit-learn, XGBoost, Isolation Forest for custom fraud models

Cloud & Big Data

- **AWS/Azure/GCP** – increasingly used for scalable data storage and ML deployment
- **Snowflake, Databricks** – modern data platforms adopted by digital-first insurers/InsurTechs

Workflow Tools

- **Git/GitHub** – version control for model code
- **Airflow** – scheduling data pipelines for recurring reports
- **Jira/Confluence** – project and documentation management

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06. Real-World Use Cases

Underwriting & Pricing

- **Risk-based Premium Pricing** – building GLM/ML models to price policies based on risk factors
- **Usage-Based Insurance (UBI)** – telematics-driven dynamic pricing for motor insurance
- **Automated Underwriting** – instant policy issuance for low-risk applicants using rules + ML models

Claims Management

- **Fraud Detection Models** – as detailed in the lifecycle example above
- **Claims Severity Prediction** – estimating expected claim payout early in the claims process for better reserving
- **Straight-Through Processing (STP)** – automating low-complexity claims for faster settlement

Customer Analytics

- **Churn/Lapse Prediction** – identifying policyholders likely to not renew
- **Cross-sell/Up-sell Models** – recommending additional coverage (e.g., health add-ons to life insurance customers)
- **Customer Segmentation** – grouping policyholders for targeted product design and marketing

Actuarial & Risk

- **Reserve Estimation (Loss Triangles)** – predicting future claim liabilities using historical development patterns
- **Catastrophe Modeling** – estimating potential losses from natural disasters for reinsurance planning
- **Mortality/Morbidity Modeling** – core life/health insurance actuarial work

Distribution & Operations

- **Agent Performance Analytics** – identifying high-performing agents/brokers and areas needing support
- **Call Center Analytics** – analyzing customer service interactions to improve satisfaction and reduce complaints

Example Interview-Style Problem Statements

- *"Build a model to predict which health insurance claims are likely to be fraudulent using claim and provider data."*
- *"How would you design a usage-based motor insurance pricing model using telematics data?"*
- *"Our life insurance persistency ratio has dropped — how would you investigate and build a model to predict lapses?"*
- *"Estimate the reserves needed for a line of business using historical claims development data."*

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07. ML & Statistical Techniques

Pricing & Underwriting Techniques

- **Generalized Linear Models (GLMs)** – the actuarial industry standard for pricing (Poisson for frequency, Gamma for severity)
- **Generalized Additive Models (GAMs)** – capturing non-linear relationships while retaining interpretability
- **Gradient Boosting (XGBoost, LightGBM)** – increasingly used for pricing where regulatory constraints allow more complex models

Fraud Detection Techniques

- **Logistic Regression, Random Forest, XGBoost** – supervised fraud classification
- **Isolation Forest, Autoencoders** – unsupervised anomaly detection for novel fraud patterns
- **Graph/Network Analysis** – detecting organized fraud rings through shared entities (phone numbers, addresses, garages)

- **Semi-supervised Learning** – addressing the biased-labeling problem where only investigated claims have confirmed labels

Actuarial-Specific Techniques

- **Chain Ladder Method / Bornhuetter-Ferguson Method** – classical actuarial reserving techniques using loss development triangles
- **Survival Analysis** – for mortality/lapse modeling in life insurance
- **Credibility Theory** – blending individual and group risk experience for more stable pricing estimates

Customer Analytics Techniques

- **Logistic Regression / Survival Analysis** – for churn/lapse prediction
- **Clustering (K-Means)** – for customer segmentation
- **Collaborative Filtering / Association Rules** – for cross-sell recommendation engines

Catastrophe & Risk Modeling

- **Extreme Value Theory (EVT)** – modeling tail risk of catastrophic events
- **Monte Carlo Simulation** – simulating catastrophe scenarios for reinsurance capital planning

Model Validation Concepts

- **Gini Coefficient / Lift Charts** – standard for evaluating pricing model discriminatory power
- **A/B Testing** – testing new pricing/underwriting rules before full rollout
- **Precision-Recall AUC** – preferred over ROC-AUC for rare-event fraud models

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08. Resources

Public Datasets for Practice

- **Kaggle: "Porto Seguro's Safe Driver Prediction"** – motor insurance claim prediction dataset
- **Kaggle: "Allstate Claims Severity"** – predicting claim severity for auto insurance
- **Kaggle: "Health Insurance Cross Sell Prediction"** – cross-sell propensity modeling dataset
- **French Motor Third-Party Liability dataset (CASdatasets in R)** – classic actuarial pricing dataset
- **Kaggle: "IBNR/Loss Reserving" datasets** – for practicing actuarial reserving techniques

Recommended Courses

- Loss Data Analytics (free online actuarial textbook/course by actuarial societies)
- Society of Actuaries (SOA) / Institute and Faculty of Actuaries (IFoA) exam study material — deep actuarial foundations
- Insurance Analytics courses on Coursera/edX
- GLM for Insurance Pricing courses (various providers, including Willis Towers Watson training)

Books

- *"Loss Models: From Data to Decisions"* – Klugman, Panjer, Willmot (actuarial modeling standard reference)
- *"Predictive Modeling Applications in Actuarial Science"* – Frees, Derrig, Meyers
- *"Fraud Analytics Using Descriptive, Predictive, and Social Network Techniques"* – Bart Baesens

Communities & Blogs

- Society of Actuaries (SOA.org) publications and research
- Actuarial Outpost forums
- Kaggle competition discussions on insurance datasets
- LinkedIn groups: "Insurance Analytics", "Actuarial Data Science"

Regulatory Reference Material

- IRDAI Annual Reports and Circulars (India) – irdai.gov.in
- NAIC Model Laws and Regulations (US) – naic.org
- EIOPA Solvency II guidelines (EU)

GitHub Repositories for Reference

- Search GitHub for: insurance-claim-prediction, actuarial-reserving, fraud-detection-insurance for open-source implementation examples

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